

Ghosh's monic octic equation has roots expressible in radicals

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Abstract

In this paper, I describe the roots of my monic octic equation, which are expressible in radicals.
The paper ends with The End

Introduction

In a previous paper, I've described one of the roots of the quartic equation and how that can be used to solve the general quartic equation in radicals.

In a previous paper, I've described my monic octic. In this paper, I describe the roots of my monic octic equation, which are expressible in radicals.

Ghosh's monic octic equation has roots expressible in radicals

When

$$Q \neq 0$$

by the right hand side of Ghosh's monic octic identity,

Ghosh's monic octic equation

$$\left(x^4 + (a - P)x^3 + Qx^2 + Qx + Q\right) \left(x^4 + Px^3 + \frac{(f - g)}{Q}x^2 + \frac{(g - h)}{Q}x + \frac{h}{Q}\right) = 0$$

can be written as the product of a quartic equation and a quartic equation, both of which can be solved in radicals.

The End